



BRNO UNIVERSITY OF TECHNOLOGY

VYSOKÉ UČENÍ TECHNICKÉ V BRNĚ

FACULTY OF INFORMATION TECHNOLOGY

FAKULTA INFORMAČNÍCH TECHNOLOGIÍ

DEPARTMENT OF INTELLIGENT SYSTEMS

ÚSTAV INTELIGENTNÍCH SYSTÉMŮ

GCC PLUGIN FOR STATIC ANALYZER SUPPORT

GCC PLUGIN PRO STATICKÉ ANALYZÁTORY

BACHELOR'S THESIS

BAKALÁŘSKÁ PRÁCE

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Bachelor's Thesis Assignment



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Institut: Department of Intelligent Systems (DITS)
Student: **Pšeja Lukáš**
Programme: Information Technology
Title: **GCC plugin for static analyzer support**
Category: Software analysis and testing
Academic year: 2025/26

Assignment:

1. Get acquainted with *CodeListener* component from the *Predator* project. Analyze the implementation of *CodeListener* and similar projects. Additionally, study the relevant data structure visualization concepts.
2. Design a new GCC plugin as a modern alternative to *CodeListener* and add new functionality (built-in visualization support and export in JSON format). Focus on improvement of the plugin interface.
3. Implement the plugin in C++ along with appropriate examples for demonstration purposes. Verify the plugin's compatibility by using it with *Predator*.
4. Test the plugin using the created examples with various GCC versions. Evaluate obtained results.

Literature:

- GitHub repository of Predator project.
<https://github.com/kdudka/predator>

Requirements for the semestral defence:

- First two points of assignment.

Detailed formal requirements can be found at <https://www.fit.vut.cz/study/theses/>

Supervisor: **Peringer Petr, Dr. Ing.**
Head of Department: Kočí Radek, Ing., Ph.D.
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Abstract

This bachelor's thesis deals with the analysis of the Code Listener infrastructure, which serves as the foundation for creating static analyzers, and its subsequent refactoring into a new plugin.

Abstrakt

Tato bakalářská práce se zabývá analýzou infrastruktury Code Listener, která poskytuje základ pro tvorbu statických analyzátorů zdrojového kódu, a její následnou refaktorizací do nového pluginu.

Keywords

Code Listener infrastructure, GCC, LLVM, plugin, refactoring, static analysis

Klíčová slova

infrastruktura Code Listener, GCC, LLVM, plugin, refaktorování, statická analýza

Reference

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GCC plugin for static analyzer support

Declaration

I hereby declare that this Bachelor's thesis was prepared as an original work by the author under the supervision of Dr. Ing. Petr Peringer. I have listed all the literary sources, publications and other sources, which were used during the preparation of this thesis.

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Lukáš Pšeja
November 27, 2025

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Contents

1	Introduction	3
2	Preliminaries	4
3	Design	5
4	Implementation	6
5	Testing	7
6	Conclusion	8
	Bibliography	9

List of Figures

Chapter 1

Introduction

With the ever-increasing complexity of software systems and the growing demand for high-quality and secure code, formal verification methods have become essential in some parts of the software development industry [?]. Formal verification methods are a set of mathematical techniques used to prove the correctness of software systems, going beyond traditional testing and debugging methods, which only test a finite number of inputs, allowing potentially critical errors to go undetected [?]. The implementation of such formal methods is not straightforward, as it requires a deep understanding of both the verified software and the application of formal methods, typically necessitating specialized domain knowledge. Formal verification tools, such as Predator, utilize static analysis to verify code written by developers. In Predator’s case, this involves verifying sequential C programs for memory safety while operating with pointers and linked lists [?]. Predator is built on top of the Code Listener infrastructure [?], which provides an application programming interface (API) foundation for creating such static analyzers. However, the majority of the Code Listener infrastructure was written nearly fifteen years ago, which has led to certain challenges in continuing its development and maintenance in its current form. To address these challenges, this bachelor’s thesis focuses on refactoring the Code Listener infrastructure into a modern GCC [?] plugin.

Chapter 2

Preliminaries

preliminaries

Chapter 3

Design

design

Chapter 4

Implementation

implementation

Chapter 5

Testing

testing

Chapter 6

Conclusion

conclusion

Bibliography